TSM 099

Homework 7

Rational equations and excluded values

Topics 29 | Estimated time: about 1 hour

Complete all problems and show your work on separate paper. Plan for about 45-50 minutes of work and 10-15 minutes to check, correct, and submit.

Use available resources to check your answers and reasoning. Complete the self-evaluation in Blackboard, upload a clear photo of your one best representative page, and retain all your written work.

Work without a calculator before checking. Give exact answers. For equations with fractions, combine and simplify each side first. Then move each remaining denominator to multiply the entire opposite side. Check solutions in the original equation. State every original denominator restriction before simplifying.

Simplify each side, then move denominators

1. State restrictions, solve, and check in the original equation.

$$\frac{24}{3x-6}+\frac{2}{x-2}=5$$

2. State restrictions, solve, and check in the original equation.

$$\frac{3}{x+1}=\frac{2}{x-2}$$

3. State restrictions, solve, and check in the original equation.

$$\frac{x+2}{x-1}=3$$

4. State restrictions, solve, and check in the original equation.

$$\frac{1}{x}+\frac{1}{x+1}=\frac{3}{x(x+1)}$$

5. State restrictions, solve, and check every candidate.

$$\frac{2}{x-1}=\frac{x}{x-1}$$

6. State restrictions, solve, and check every candidate.

$$\frac{x}{x-2}-\frac{2}{x-2}=2$$

Keep restrictions through cancellation

7. State restrictions, simplify both sides, and give the full solution set.

$$\frac{x+1}{x-1}=1+\frac{2}{x-1}$$

8. State restrictions, solve, and check every candidate.

$$\frac{x}{x-1}=\frac{1}{x-1}$$

9. State restrictions, solve, and check every candidate.

$$\frac{1}{x-1}+\frac{1}{x+1}=1$$

10. A student turns 1/x + 2 = 3/x into 1 + 2 = 3 by moving denominators before combining whole sides. Explain the error, solve correctly, and check.